

Experiment 9 – Automobile Speed Control using ANFIS Hybrid System

1. Theory

- **ANFIS (Adaptive Neuro-Fuzzy Inference System)** merges fuzzy logic and neural networks.
- Fuzzy rules express human-like reasoning (“If slope ↑ and speed ↓ → increase throttle”).
- Neural learning adjusts membership functions and rule weights automatically.
- The system learns input–output relations from data to maintain vehicle speed under varying road slopes and loads.
- Uses a **Sugeno-type FIS** trained by a **hybrid algorithm** (least-squares + back-propagation).

2. Working (Detailed)

1. Inputs (Slope Angle, Throttle %)

- Represent external driving conditions and driver’s torque demand.
- Provide real-time data that influence vehicle acceleration and control response.

2. Error (e) and Change in Error (de)

- Error = (Desired Speed – Actual Speed); Change in Error shows how fast the error is varying.
- These values form the control feedback that guides throttle adjustments.

3. Scaling with Gains (Ke, Kde, Ku)

- Ke and Kde normalize the input signals to fit within fuzzy inference limits.
- Ku scales the final output signal to suitable actuator range for throttle/brake control.

Symbol	Full Form	Acts On	Role	Formula
Ke	Error Gain	Error (e)	Normalizes speed difference	$e(\text{scaled}) = K_e \times e$
Kde	Derivative Gain	Change of error (de/dt)	Normalizes rate of error change	$de(\text{scaled}) = K_{de} \times de/dt$
Ku	Output Gain	Control signal (u)	Denormalizes fuzzy output	$u(\text{final}) = K_u \times u(\text{fuzzy})$

4. ANFIS Controller Core (Fuzzy + Neural Integration)

- Fuzzy logic layer interprets linguistic rules such as “If error is small and decreasing → reduce throttle.”
- Neural learning layer continuously tunes membership functions and rule weights for optimal behavior.

5. ACO Optimization Loop (Metaheuristic Layer)

- Ant Colony Optimization fine-tunes K_e , K_{de} , K_u , λ , μ to minimize control error metrics (IAE, ISE, RMSE).
- Ants explore multiple parameter paths and reinforce best-performing combinations through pheromone updating.

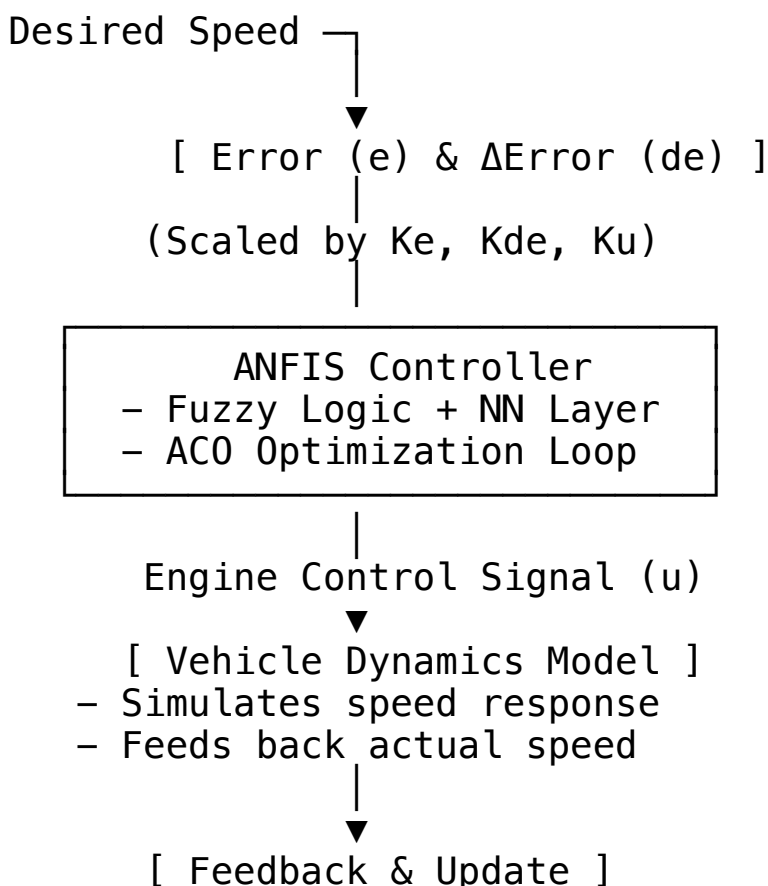
6. Vehicle Dynamics Block (Plant Model)

- Receives control signal $u(t)$ and converts it into vehicle speed response based on inertia and load.
- Provides nonlinear simulation of real-world conditions like slope variation and friction.

7. Feedback and Convergence

- The actual speed is continuously fed back to recalculate the new error.
- Once minimal error and steady-state stability are achieved, final optimal parameters are fixed for the ANFIS controller.

3. Block Diagram (with Explanation)



- Error recalculated
 - Parameters adjusted
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4. Advantages

- Automatically adapts to nonlinear and varying road conditions.
 - Requires little manual tuning once trained.
 - Provides smooth and stable speed response.
 - Combines interpretability of fuzzy logic with learning power of neural nets.
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5. Disadvantages

- High computational cost during training.
 - Performance depends on quality of training data.
 - Choosing initial membership functions is critical.
 - Real-time hardware implementation may be resource-intensive.
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